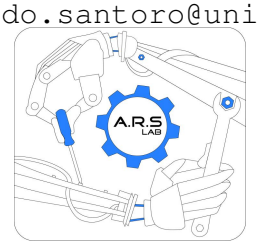
A Recap on Dynamic Systems

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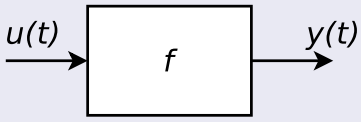
Advanced Robotics and Autonomous Systems

Notes

A System

Each **real system** (any object belonging to the real world) is an object with finite number of freedom degrees that evolves in time according to a certain law

A real system can be represented as a **black box** that can be externally stimulated from an input (that we call $u(t)$), producing a certan **effect** which is the output (that we call $y(t)$)



$$y(t) = f(u(t))$$

The behaviour is **totally represented** by function $f(\cdot)$

Notes

Definition of **Dynamic System**

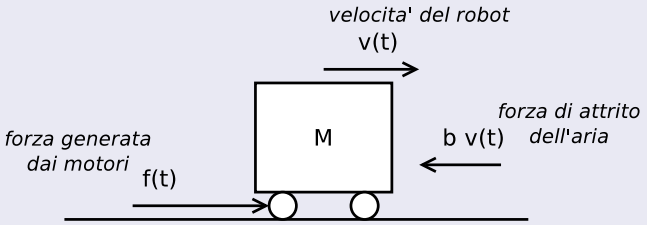
A **dynamic system** is a (physical) system where, given a certain time instant t, the output $y(t)$ depends on the current value $u(t)$ and the past of $u(t)$ and $y(t)$

A dynamic system (**time-continuous**) is described by a **differential equation** in the time domain:

$$y = f(\dot{y}, \ddot{y}, ..., u, \dot{u}, \ddot{u}, t)$$

Notes

The Cart as a **Dynamic System**



Input = $f(t)$ *Output* =$p(t)$

Let's consider again the cart. If we use the force $f(t)$ as the input and the position $p(t)$ as the output, we can model the cart using the following differential equation:

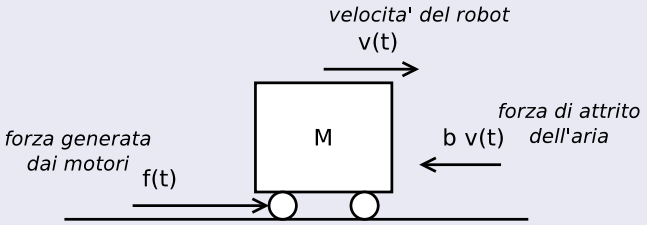
$$f - b\,\dot{p} = M\,\ddot{p}$$

(recall that $v = \dot{p}$ and $a = \ddot{p}$)

It is a second-order differential equation, that, at first sight, is hard to solve and also to simulate

Notes

The Cart as a **Dynamic System**



Input = $f(t)$ *Output* =$p(t)$

$$f - b\,\dot{p} = M\,\ddot{p}$$

But if we introduce also the v variable, we can rewrite the equation above as a **system** of differential equation:

$$\begin{cases} \dot{v} &= -\frac{b}{M}v + \frac{1}{M}f \\ \dot{p} &= v \end{cases}$$

Here we have only first-order differential equations that can be handled more easily

Notes

Representation of a **Dynamic System**

A dynamic system described by a n-order **differential equation** as:

$$y = f(\dot{y}, \ddot{y}, ..., u, \dot{u}, \ddot{u}, t)$$

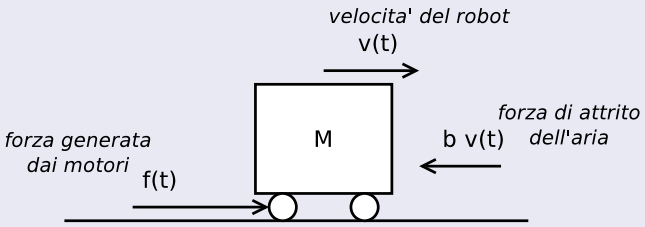
can be represented as a **system of n differential equations** of the first-order by using additional variables that are equal to the first-derivative, second-derivative, etc.

Variables that are derivated are called **state variables** and represent the instantaneous condition of the system

Notes

The Cart as a **Dynamic System**

State Variables



Input = $f(t)$ *Output* =$p(t)$

$$\begin{cases} \dot{v} &= -\frac{b}{M}v + \frac{1}{M}f \\ \dot{p} &= v \end{cases}$$

Here p and v are the **state variables** of our cart and the equations above that describe the evolution of such state variables are called **state equations**

Notes

Linear Systems

A **linear system** is a system described by a set of first-order linear differential equations

A **linear differential equation** is a differential equation in which a derivated variable **lineary depends** on the variable itself and other variables

$$\begin{cases} \dot{x}_1 &= a_{11}x_1 + ... + a_{1n}x_n \\ ... &... &... \\ \dot{x}_n &= a_{n1}x_1 + ... + a_{nn}x_n \end{cases}$$

$$\begin{cases} \dot{v} &= -\frac{b}{M}v + \frac{1}{M}f \\ \dot{p} &= v \end{cases}$$

Our cart model is a **linear system** with p and v as state variables

Notes

Inside a linear system

Parts of a linear system model

Given a certain system represented by a linear model, we can identify, in each equations, two basic parts:

- Terms that give dependency of other state variables (in red)
- Terms that give the dependency of the input(s) (in blue)

$$\begin{cases} \dot{v} &= -\frac{b}{M}v + \frac{1}{M}f \\ \dot{p} &= v \end{cases}$$

- Terms in red represent the **free evolution** of the system (i.e. the evolution without any input)
- Terms in blue represent the **forced evolution** of the system

Notes

Standard Representation of Dynamic Systems

Notes

Standard Representation of a Dynamic System

The Cart

- **State Equations** $\begin{cases} \dot{v} &= -\frac{b}{M}v + \frac{1}{M}f \\ \dot{p} &= v \end{cases}$
- We know, from algebra, that a set of linear equations can be represented in a **matrix form**
- Let's incapsulate state variables in a (algebraic) **vector**:

$$x = \begin{bmatrix} v \\ p \end{bmatrix} \qquad \dot{x} = \begin{bmatrix} \dot{v} \\ \dot{p} \end{bmatrix}$$

State equations in matrix form

$$\begin{bmatrix} \dot{v} \\ \dot{p} \end{bmatrix} = \begin{bmatrix} -\frac{b}{M} & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} v \\ p \end{bmatrix} + \begin{bmatrix} \frac{1}{M} \\ 0 \end{bmatrix} f$$

$$\dot{x} = \begin{bmatrix} -\frac{b}{M} & 0 \\ 1 & 0 \end{bmatrix} x + \begin{bmatrix} \frac{1}{M} \\ 0 \end{bmatrix} f$$

Notes

Standard Representation of a Dynamic System

State Equation in matrix form

$$\begin{bmatrix} \dot{v} \\ \dot{p} \end{bmatrix} = \begin{bmatrix} -\frac{b}{M} & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} v \\ p \end{bmatrix} + \begin{bmatrix} \frac{1}{M} \\ 0 \end{bmatrix} f$$

$$\dot{x} = \begin{bmatrix} -\frac{b}{M} & 0 \\ 1 & 0 \end{bmatrix} x + \begin{bmatrix} \frac{1}{M} \\ 0 \end{bmatrix} f$$

Let:

$$A = \begin{bmatrix} -\frac{b}{M} & 0 \\ 1 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} \frac{1}{M} \\ 0 \end{bmatrix} \qquad u = f$$

We obtain the **general state equation** in **matrix form**:

$$\dot{x} = A\ x + B\ u$$

Notes

Standard Representation of a Dynamic System

Output Equation in matrix form

- The **output** of our system y is (in general) a linear combination of state variables (state vector) $x = \begin{bmatrix} v \\ p \end{bmatrix}$
- If we consider the position p as output, to obtain it we can simply multiply the state vector with $\begin{bmatrix} 0 & 1 \end{bmatrix}$

$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ p \end{bmatrix}$$

Generalising we have:

$$y = C\,x$$

$$C = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

Notes

Standard Representation of a Linear Dynamic System

In summary, a **linear system time-continuous** is totally specified by the following matrix equations:

$$\begin{cases} \dot{x} &= Ax + Bu \\ y &= Cx \end{cases}$$

- x, state vector (n elements)
- u, input vector (m elements)
- y, output vector (p elements)
- A, **state matrix** $n \times n$
- B, **input matrix** $n \times m$
- C, **output matrix** $p \times n$

Notes

Roles of Matrices in the Matrix Form

$$\dot{x} = A\,x + B\,u \tag{1}$$

The state equation has two parts:

- Matrix A characterises the **evolution of the state**
- Matrix B represents the contribution of the input
- When $u = 0$, we have the **free evolution** and the state equation becomes: $\dot{x} = A\,x$
- The **dynamics (behaviour)** of the system is totally specified by matrix A
- In other words, if we have a representation in the form (1) we do not need to solve the differential equations or perform a simulation to understand the behaviour of the system, but it is sufficient to analyse matrix A

Notes

Discretization of a Linear System Model

Notes

Discretization of a Linear System

- In order to handle/implement a linear system (given its model in matrix form), we can consider the sampling of any variable with prefixed ΔT
- The derivative becomes **incremental ratio** (first-order approximation):

$$
\begin{aligned}
x(t), t \in \mathcal{R} &\Rightarrow x(k\Delta T), k \in \mathcal{N} \\
\dot{x}(t), t \in \mathcal{R} &\Rightarrow \frac{x((k+1)\Delta T) - x(k\Delta T)}{\Delta T}, k \in \mathcal{N} \\
x(t) &\Rightarrow x(k) \\
\dot{x}(t) &\Rightarrow \frac{x(k+1) - x(k)}{\Delta T}
\end{aligned}
$$

Notes

Discretization of a Linear System

The system

$$\begin{cases} \dot{x} &= Ax + Bu \\ y &= Cx \end{cases}$$

discretized using a sampling interval ΔT becomes:

$$\begin{cases} \frac{x(k+1)-x(k)}{\Delta T} &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) \end{cases}$$

$$\begin{cases} x(k+1) - x(k) &= A\Delta Tx(k) + B\Delta Tu(k) \\ y(k) &= Cx(k) \end{cases}$$

$$\begin{cases} x(k+1) &= A\Delta Tx(k) + x(k) + B\Delta Tu(k) \\ y(k) &= Cx(k) \end{cases}$$

$$\begin{cases} x(k+1) &= (A\Delta T + I)x(k) + B\Delta Tu(k) \\ y(k) &= Cx(k) \end{cases}$$

Notes

Discretization of a Linear System

The system:

$$\begin{cases} \dot{x} &= Ax + Bu \\ y &= Cx \end{cases}$$

becomes:

$$\begin{cases} x(k+1) &= \widetilde{A}\, x(k) + \widetilde{B}\, u(k) \\ y(k) &= \widetilde{C}\, x(k) \end{cases}$$

here:

$$\begin{aligned} \widetilde{A} &= A\Delta T + I \\ \widetilde{B} &= B\Delta T \\ \widetilde{C} &= C \end{aligned}$$

Notes

Standard Model from Linear Differential Equation

Notes

Standard Model from Linear Differential Equation

Standard Model from Linear Differential Equation

Let us suppose we derived a model of a system given by:

$$\ddot{y} + 3\dot{y} + 0.5y = 4u$$

How can we derive a standard representation in the form $x = Ax + Bu$?

Rules (by example)

1. We have a second derivative, so the system will be of the **second order**
2. The variable y is differentiated (two times) so our state variables come from it, and will be its derivatives
3. We apply a variable change in order to have equations with **only first derivative**

Notes

Standard Model from Linear Differential Equation

Standard Model from Linear Differential Equation

$$\ddot{y} + 3\dot{y} + 0.5y = 4u$$

Rules (by example)

- Let's assume:

$$\begin{aligned} x_1 &= y \\ x_2 &= \dot{y} \end{aligned}$$

- We have:

$$\begin{aligned} x_1 &= y \\ x_2 &= \dot{y} = \dot{x_1} \\ \dot{x_2} &= \ddot{y} \end{aligned}$$

Notes

Standard Model from Linear Differential Equation

Standard Model from Linear Differential Equation

$$\ddot{y} + 3\dot{y} + 0.5y = 4u$$

Rules (by example)

$$\begin{aligned} x_1 &= y \\ x_2 &= \dot{y} = \dot{x}_1 \\ \dot{x}_2 &= \ddot{y} \end{aligned}$$

Let's substitute

$$\begin{aligned} \dot{x}_2 + 3x_2 + 0.5x_1 &= 4u \\ x_2 &= \dot{x}_1 \end{aligned}$$

Notes

Standard Model from Linear Differential Equation

Standard Model from Linear Differential Equation

$$\ddot{y} + 3\dot{y} + 0.5y = 4u$$

Let's reorder

$$\begin{aligned} \dot{x_1} &= x_2 \\ \dot{x_2} &= -0.5x_1 - 3x_2 + 4u \end{aligned}$$

And put system sign:

$$\begin{cases} \dot{x_1} &= x_2 \\ \dot{x_2} &= -0.5x_1 - 3x_2 + 4u \end{cases}$$

Notes

Standard Model from Linear Differential Equation

Let's put in standard matrix form

$$\begin{cases} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -0.5x_1 - 3x_2 + 4u \end{cases}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad A = \begin{bmatrix} 0 & 1 \\ -0.5 & -3 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 4 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

So we have:

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx \end{aligned}$$

Notes

Standard Model from Linear Differential Equation

Another Example

$$\dddot{y} + 4\ddot{y} + 2\dot{y} + y = 3u$$

Rules (by example)

- Let's assume:

$$\begin{aligned} x_1 &= y \\ x_2 &= \dot{y} = \dot{x}_1 \\ x_3 &= \ddot{y} = \dot{x}_2 \\ \dot{x}_3 &= \dddot{y} \end{aligned}$$

- We have:

$$\dot{x}_3 + 4x_3 + 2x_2 + x_1 = 3u$$

Notes

Standard Model from Linear Differential Equation

Another Example

$$\dddot{y} + 4\ddot{y} + 2\dot{y} + y = 3u$$

Let's substitute

$$\begin{aligned} \dot{x_3} + 4x_3 + 2x_2 + x_1 &= 3u \\ x_2 &= \dot{x_1} \\ x_3 &= \dot{x_2} \end{aligned}$$

Notes

Standard Model from Linear Differential Equation

Let's reorder and put in standard matrix form

$$\begin{cases} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ \dot{x}_3 &= -x_1 - 2x_2 - 4x_3 + 3u \end{cases}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -4 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

So we have:

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx \end{aligned}$$

Notes

Equilibrium

Notes

Equilibrium

Generic Definition of Equilibrium state

Given a dynamic system defined by state equation $\dot{x} = f(x, u)$, with $u = u(t)$ as input, the state $\overline{x} = x(T^*)$ is an **equilibrium** if

$$f(\overline{x}, u(t)) = 0, \forall t \geq T^*$$

that is, no state variations, thus $\dot{x} = 0$

In other words, an equilibrium state means no further state variations for $t \geq T^*$

Notes

Equilibrium

Equilibrium in Continuous Linear Systems

Given a linear system defined by the state equation

$$\dot{x} = Ax + Bu$$

the state $\overline{x}$ is an **equilibrium** if

$$0 = A\overline{x} + Bu$$

Searching for equilibrium states in a linear system implies to solve the (linear) equation:

$$x = -A^{-1}Bu$$

if A is *not invertible* then there are either **no equilibrium states** or **infinite equilibrium states**

Notes

Equilibrium

Equilibrium in Discrete Linear Systems

Given a linear system defined by the state equation

$$x(k+1) = \widetilde{A}\, x(k) + \widetilde{B}\, u$$

the state $\overline{x}$ is an **equilibrium** if

$$\overline{x} = \widetilde{A}\, \overline{x} + \widetilde{B}\, u$$

Searching for equilibrium states in a linear system implies to solve the (linear) equation:

$$x = -(A - I)^{-1} B u$$

if $A - I$ is *not invertible* then there are either **no equilibrium states** or **infinite equilibrium states**

Notes

Equilibrium

Equilibrium in the Cart

Let's consider the cart: $\begin{cases} \dot{v} &= -\frac{b}{M}v + \frac{1}{M}f \\ \dot{p} &= v \end{cases}$

$$\begin{bmatrix} \dot{v} \\ \dot{p} \end{bmatrix} = \begin{bmatrix} -\frac{b}{M} & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} v \\ p \end{bmatrix} + \begin{bmatrix} \frac{1}{M} \\ 0 \end{bmatrix} f$$

Let's consider a **constant** input F; here matrix A is singular so we have:

$$\begin{cases} 0 &= -\frac{b}{M}v + \frac{1}{M}F \\ 0 &= v \end{cases}$$

and:

$$\begin{cases} v &= \frac{F}{b} \\ v &= 0 \end{cases}$$

Relations above are in contrast and this comes from the singularity of A

Indeed the solution is $v = \frac{F}{b}$ and $p = any$ (so infinite equilibrium points)

Notes

Equilibrium without Input

Equilibrium in the Cart

Let's consider the cart in **free evolution**: $\begin{cases} \dot{v} &= -\frac{b}{M}v \\ \dot{p} &= v \end{cases}$

If we determine the equilibrium points we have:

$$\begin{cases} v &= 0 \\ p &= any \end{cases}$$

Indeed, if we have **no inputs**, there are **infinite** equilibrium points where the speed is 0

Notes

Stability

Notes

Stability

Simple Stability

An equilibrium point $x_0 = x(T^*)$ is **stable**, when, after a perturbation of the state at time $\bar{t} \geq T^*$, the resulting state evolution (trajectory) is **always near** point $x_0, \forall t \geq \bar{t}$:

$$\exists M > 0 : \|x(t) - x_0\| < M, \forall t \geq \bar{t}$$

Notes

Asymptotical Stability

An equilibrium point $x_0 = x(T^*)$ is **asymptotically stable**, when, after a perturbation of the state at time $\bar{t} \geq T^*$, the resulting state evolution (trajectory) **tends to** x_0:

$$\lim_{t \to \infty} x(t) = x_0$$

Notes

Instability

An equilibrium point $x_0 = x(T^*)$ is **instable**, when, after a perturbation of the state at time $\bar{t} \geq T^*$, the resulting state evolution (trajectory) **always steps away from** x_0:

$$\lim_{t \to \infty} x(t) = \infty \quad \text{or it does not exists}$$

Notes

Stability

Stability in the Cart

The equilibrium is:

$$\begin{cases} v = 0 \\ p = any \end{cases}$$

It is **simply stable**

Given a position p_1, if perturbate it as p_2, the cart remains in p_2 thus "near" p_1

Notes

Stability of a System

Stability in a Linear System

A **linear system** is said to be **stable** if all state trajectories are **stable**

Notes

Trajectories, Stability and Eigenvalues

Notes

The Fundamental Theorem of the Algebra

Roots of a Polynomial

Given of polynomian in x with degree n with real coefficients:

$$a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n \qquad a_i \in \mathcal{R}$$

then the **roots** will be:

- either **all** real ($\in \mathcal{R}$)
- or, if some of them are complex, they will be **complex and conjugate**

Notes

The Characteristic Polynomial

Given a square matrix $A = \{a_{i,j}\} \in \mathcal{R}^{n \times n}$, then its **eigenvalues** will be:

- either **all** real ($\in \mathcal{R}$)
- or, if some of them are complex, they will be **complex and conjugate**

Let's recall that the eigenvalues of a matrix A are the roots of the characteristic polynomial:

$$|\lambda I - A|$$

Notes

Merging all the worlds

Given a linear system in free running defined by

$$\dot{x} = Ax$$

given that the eigenvalues of A are:

$$\lambda_1, \lambda_2, ..., \sigma_k \pm i\omega_k, \sigma_{k+1} \pm i\omega_{k+1} ...$$

with $\lambda_i \in \mathcal{R}$ and and $\sigma_k \pm i\omega_k \in \mathcal{C}$ then the trajectory will have the form:

$$x(t) = K_1 e^{\lambda_1 t} + ... + K_{k-1} e^{\lambda_{k-1} t} + e^{\sigma_k t}(K1_k \cos \omega_k t + K2_k \sin \omega_k t) ...$$

(the explaination will be given below)

Notes

Eigenvalues and Stability

Given a linear system defined by

$$\dot{x} = Ax + Bu$$

given that the eigenvalues of A are:

$$\lambda_i \in \mathcal{C} \qquad (\text{remind that } \mathcal{R} \subset \mathcal{C})$$

i.e. λ_i may be either **real** or **complex and conjugate**, we have

Condition	System Stability
All $Re(\lambda_i) < 0$	**Asymptotical**
At least $Re(\lambda_{i*}) = 0$, but, if λ_{i*} is compl-and-conj, it is simple (mutiplicity = 1)	**Simple**
All other cases	**Instability**

Notes

Link between Response and Eigenvalues

Given a square matrix $A = \{a_{i,j}\} \in \mathcal{R}^{n \times n}$ whose eigenvalues are λ_i, then the following holds:

$$A = T^{-1} E \; T$$

where:

- T is a $n \times n$ invertible matrix
- E is a $n \times n$ **diagonal** matrix whose diagonal elements are the eigenvalues of A

Notes

Link between Response and Eigenvalues

Given a linear system defined by

$$\dot{x} = Ax + Bu$$

and given that the eigenvalues of A are: λ_i, we can write:

$$\begin{aligned} \dot{x} &= T^{-1} E\ T\ x + B\ u \\ T\dot{x} &= T\ T^{-1} E\ T\ x + T\ B\ u \\ T\dot{x} &= E\ T\ x + T\ B\ u \end{aligned}$$

Let's assume that $\widetilde{x} = T\dot{x}$ and $\widetilde{B} = T\ B$, we have:

$$\dot{\widetilde{x}} = E\ \widetilde{x} + \widetilde{B}\ u$$

Notes

Link between Response and Eigenvalues

The system:

$$\dot{x} = Ax + Bu$$

and

$$\dot{\tilde{x}} = E\ \tilde{x} + \tilde{B}\ u$$

are **equivalent**, i.e. **the same**, so they feature the same response and the same stability conditions

We have only applied a **change of reference frame** for state variables defined by matrix T

Notes

Link between Response and Eigenvalues

Let's consider:

$$\dot{\tilde{x}} = E \ \tilde{x}$$

that can be rewritten as:

$$\begin{cases} \dot{\tilde{x}}_1 &= \lambda_1 \ \tilde{x}_1 \\ \dot{\tilde{x}}_2 &= \lambda_2 \ \tilde{x}_2 \\ & \dots \\ \dot{\tilde{x}}_n &= \lambda_n \ \tilde{x}_n \end{cases}$$

If $\lambda_i \in \mathcal{R}$ then each differential equation can be easily integrated as:

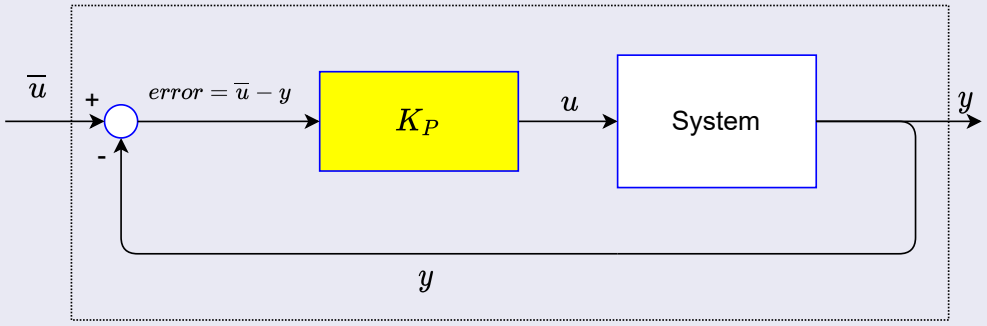
$$\tilde{x}_i \ (t) = e^{\lambda_i t}$$

C.V.D.

Notes

Dyamic Systems and Control Systems

Notes

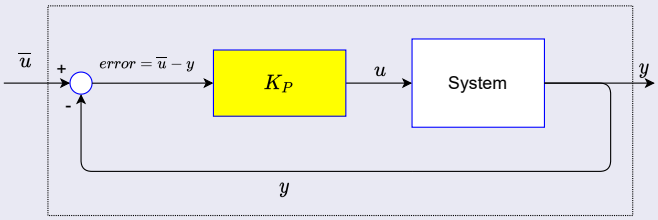


Let's consider a linear system $\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$

in feedback loop with a proportional controller

Is the resulting system a linear system as well?

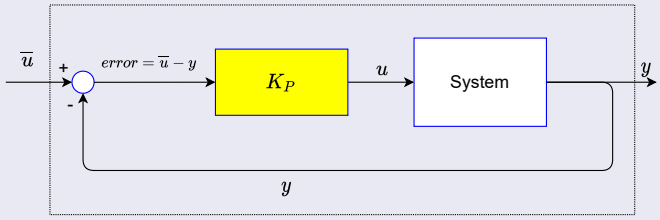
Notes



$$\begin{cases} \dot{x} &= Ax + Bu \\ y &= Cx \end{cases}$$

$$\begin{aligned} u &= K_P error \\ &= K_P(\overline{u} - y) \\ &= K_P\overline{u} - K_P y \end{aligned}$$

Notes



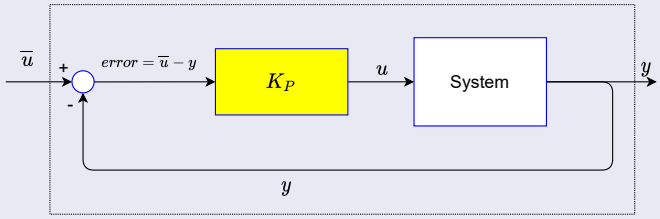
$$\begin{cases} \dot{x} &= Ax + B(K_P\overline{u} - K_P y) \\ y &= Cx \end{cases}$$

$$\begin{cases} \dot{x} &= Ax + BK_P\overline{u} - BK_P y \\ y &= Cx \end{cases}$$

$$\begin{cases} \dot{x} &= Ax + BK_P\overline{u} - BK_P Cx \\ y &= Cx \end{cases}$$

$$\begin{cases} \dot{x} &= (A - BK_P C)x + BK_P\overline{u} \\ y &= Cx \end{cases}$$

Notes



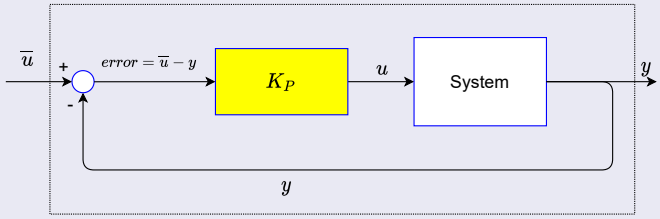
The resulting system is

$$\begin{cases} \dot{x} &= \overline{A}x + \overline{B}\,\overline{u} \\ y &= Cx \end{cases}$$

with

$$\begin{aligned} \overline{A} &= A - BK_PC \\ \overline{B} &= BK_P \end{aligned}$$

Notes



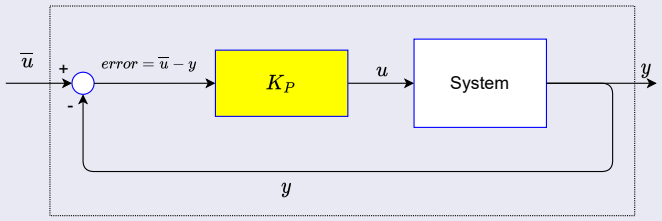
Given that:

$$\begin{aligned} \overline{A} &= A - BK_PC \\ \overline{B} &= BK_P \end{aligned}$$

we have:

- The **closed loop** system is a linear system
- The **dynamics** (behaviour) of the closed loop system depends of A, B, C, K_P (while the dependency of the orginal system were only w.r.t to A)
- The parameter K_P **strongly affects** both the dynamics and the **stability** of the closed loop system

Notes



$$\overline{A} = A - BK_PC$$
$$\overline{B} = BK_P$$

By acting on K_P we can:

- Make **instable** a stable system
- Make **stable** an instable system
- **Completely change** the system's response, e.g. by introducing or removing oscillation
- ...

Notes

Canonical Signals

Notes

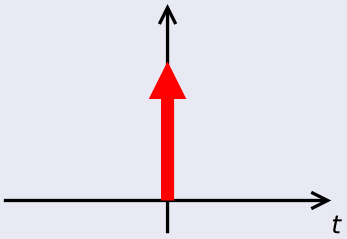
Canonical Signals

System Theory often uses some specific input signals, called **canonical signals**, to study the behaviour of a system:

- **Impulse** or **Dirac Delta**
- **(Unitary) Step**
- **Ramp**

Notes

Dirac Delta



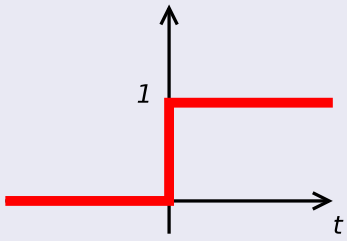
The **Dirac Delta** $\delta(t)$ is an impulsive signal that, from the mathematical point of view, is defined as:

$$\begin{cases} \delta(t) = 0, \forall t \neq 0 \\ \delta(t) = +\infty, t = 0 \\ \int_{-\infty}^{+\infty} \delta(t)dt = 1 \end{cases}$$

It is used to represent a physical fenomena with a great intensity but with an infinitesimal duration

Notes

Unitary Step



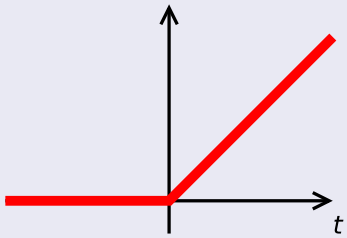
The **step** $u(t)$ is a signal defined as follows:

$$\begin{cases} u(t) = 0, \forall t < 0 \\ u(t) = 1, \forall t \geq 0 \end{cases}$$

It is used to model the application, at time 0, of a constant stimulus to a system

Notes

Ramp



The **ramp** $r(t)$ is an increasing signal defined as follows:

$$\begin{cases} r(t) & = & 0, \forall t < 0 \\ r(t) & = & t, \forall t \geq 0 \end{cases}$$

It is used to model the application to a system, at time 0, of a simulus that grows indefinitely

Notes

Relationship between Canonical Signals

Signals

$$u(t) = \int_0^t \delta(\tau) d\tau \qquad \frac{du(t)}{dt} = \delta(t)$$
$$r(t) = \int_0^t u(t) d\tau \qquad \frac{dr(t)}{dt} = u(t)$$

This is not true in classical analysis, but in terms of "theory of distributions"

Responses

- Given a **linear system**, if $y_d(t)$ is the impulse response, then the **step response** is:

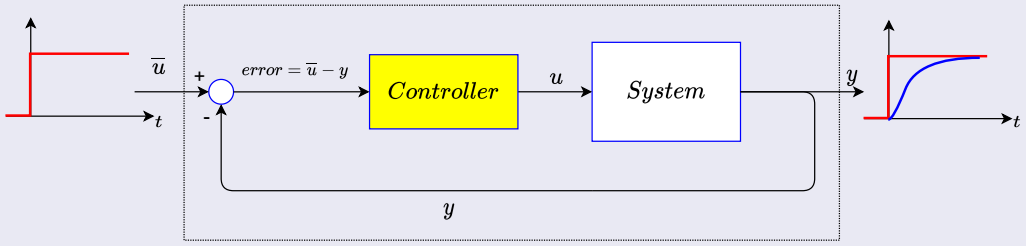
$$y_s(t) = \int_0^t y_d(\tau) d\tau$$

- and the **ramp response** is:

$$y_r(t) = \int_0^t y_s(\tau) d\tau$$

Notes

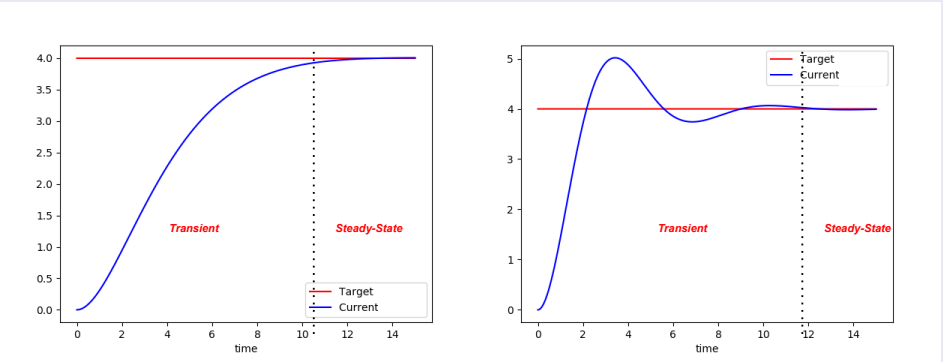
Canonical Signals and Control Systems



- Given a control system, its performaces are measured on the basis of **canonical inputs**
- The step represents a **constant reference** that is suddenly applied
- The ramp represents a **moving reference**, thus making it possible to measure the ability of the control system to follow changing references

Notes

Transient and Steady-State Regimes

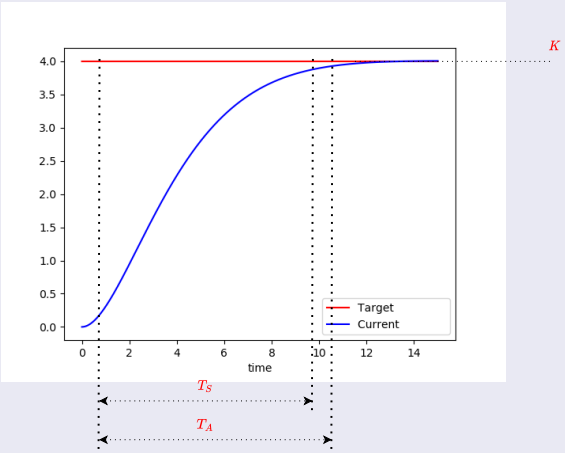


The response of a system to a step (or a pulse is composed of two parts:

- **Transient**: initial part of the response; the output changes substantially during time
- **Steady-State**: when the transient is over, the output features small or no changes and stabilise to a specific value
- According to the response type (left or right figures above), the transient features some specific characteristics

Notes

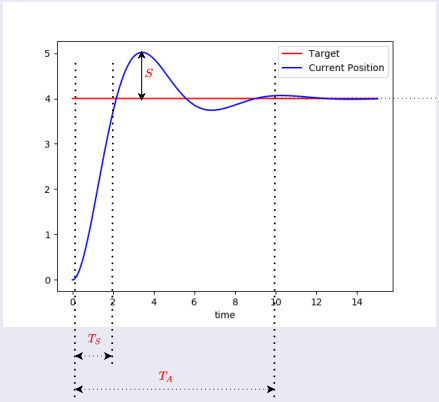
Transient Characteristics



- **Steady-State Value**: $K = \lim_{t \to \infty} y(t)$
- **Rise Time** T_S ("tempo di salita"): the time required to go from 10% of K to 90% of K
- **Set-up Time** T_A ("tempo di assestamento"): the time required to have the output around the 98% of K

Notes

Transient Characteristics



- **Steady-State Value**: $K = \lim_{t\to\infty} y(t)$
- **Rise Time** T_S ("tempo di salita"): the time required to go from 10% of K to 90% of K
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- **Overshot** S ("sovraelongazione"): the percentage w.r.t. K of the first peak $S = \frac{peak - K}{K}$

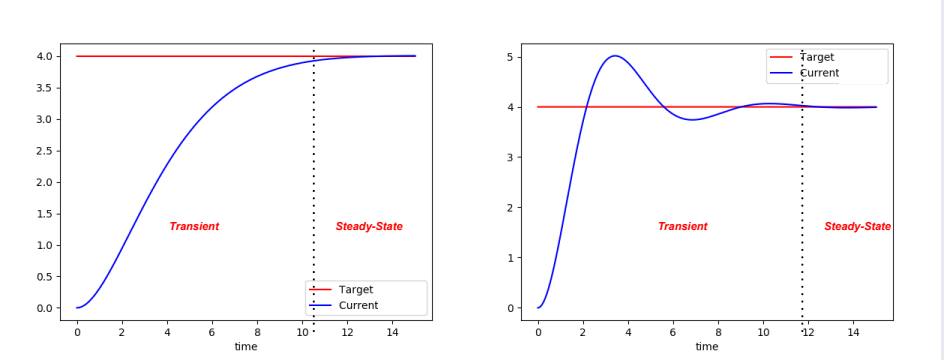
Notes

System Response and Eigenvalues

Notes

System Reponse and Eigenvalues

- Eigenvalues of state matrix A not only are able to determine system stability but also **encode** important information about the transient



- Left-side response represents a system with **real and negative** eigenvalues
- Right-side response represents a system with at least a couple of **complex and conjugate** eigenvalues with **negative real part**

Notes

System Reponse and Eigenvalues

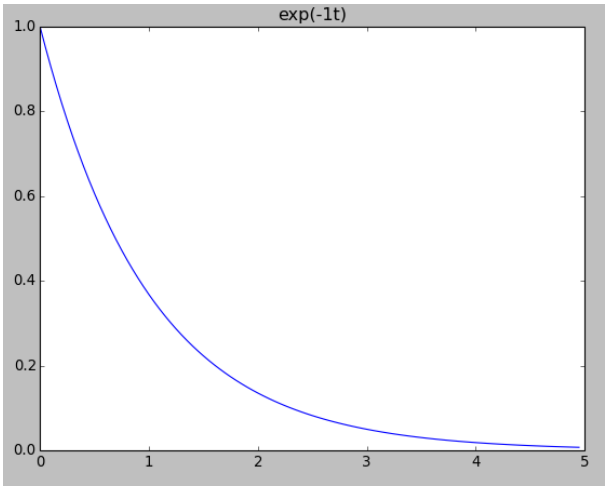
Real Negative Eigenvalues

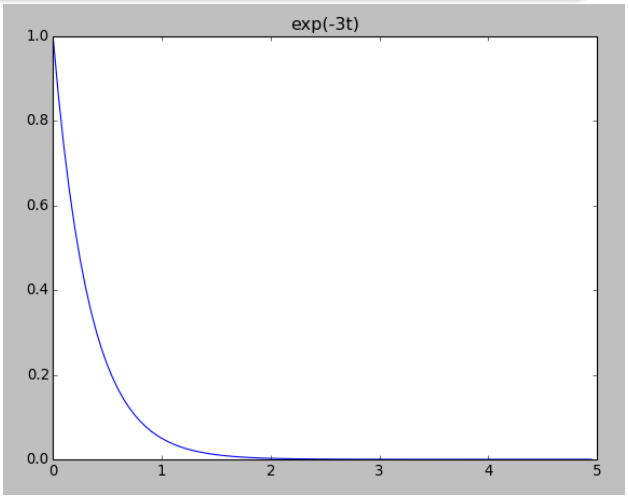
Let us consider a system with $\lambda_1 = -1$ e $\lambda_2 = -3$, then the response will be of type:

$$e^{-t} + e^{-3t}$$

From the plots we see that the duration of the transient for λ_1 is greater than the one of λ_2

In other words, λ_1 is "slower" than λ_2





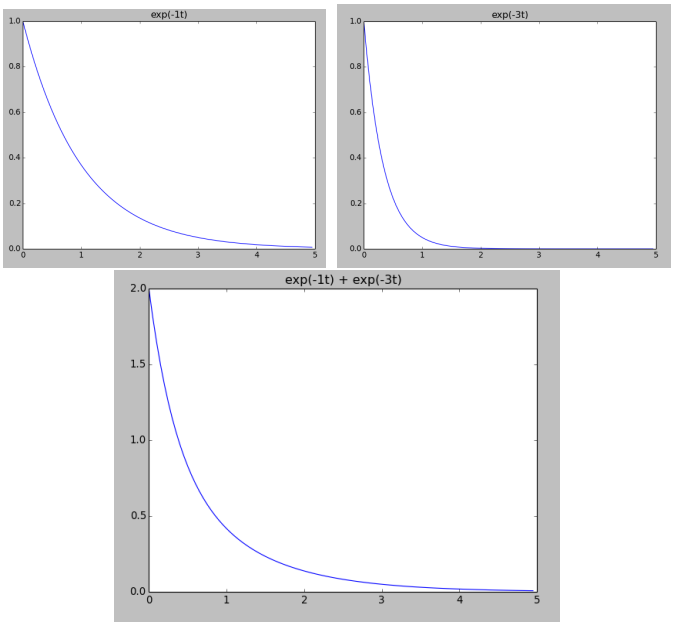
Notes

System Reponse and Eigenvalues

Real Negative Eigenvalues

If we combine the plots and see the complete response, we observe that its transient is more influenced by λ_1 rather than λ_2

In other words, the overall response is as “slow” as the one with only λ_1



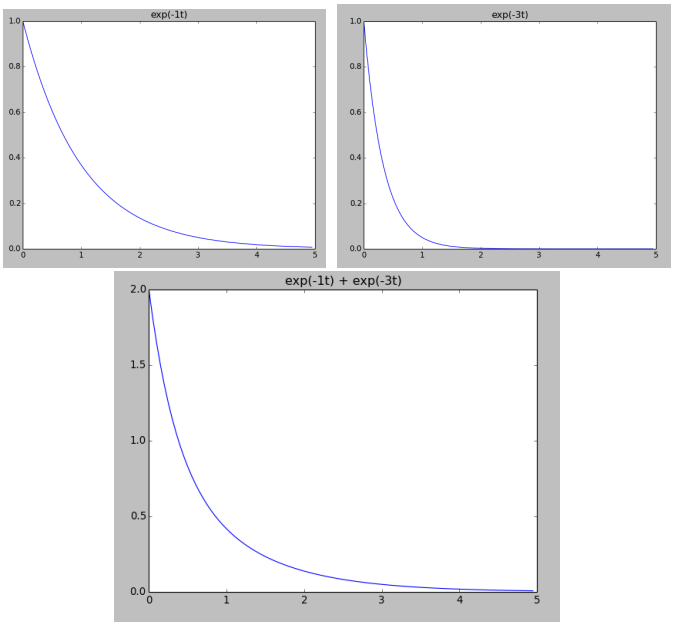
Notes

System Reponse and Eigenvalues

Dominant Eigenvalues

In other words, the overall response is as “slow” as the one with only λ_1

We say that λ_1 **dominates** λ_2, or that λ_1 is a **dominant eigenvalue**



Notes

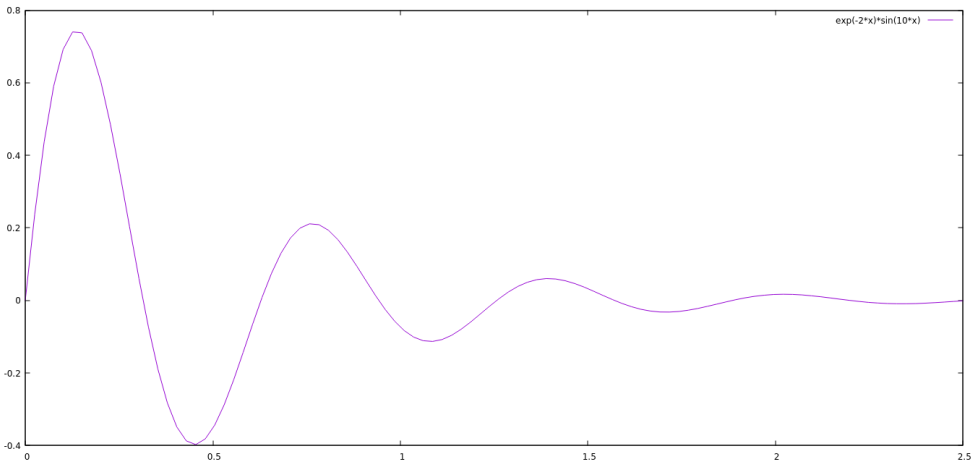
System Reponse and Eigenvalues

Complex and Conjugate Eigenvalues

If we have a couple of complex and conjugate eigenvalue, i.e. $\sigma \pm i\omega$ the response has the form

$$e^{\sigma t} \cos \omega t$$

and the value of σ (the real part) characterised the duration of the transient



Notes

System Reponse and Eigenvalues

Dominant Eigenvalue

Let's consider a asymptotically stable system with eigenvalue:

$$\lambda_1, \lambda_2, ..., \sigma_1 \pm i\omega_1, \sigma_2 \pm i\omega_2, ...$$

The **dominant eigenvalue** is $\lambda^* = \max(\lambda_1, \lambda_2, ..., \sigma_1, \sigma_2, ...)$, i.e. the highest value of the real parts

Notes

Natural Frequencies or Modes

Let's consider a asymptotically stable system with eigenvalue:

$$\lambda_1, \lambda_2, ..., \sigma_1 \pm i\omega_1, \sigma_2 \pm i\omega_2, ...$$

The real parts contribute to the response as an exponential term:

$$e^{\lambda_i t} \qquad e^{\sigma_i t}$$

The real part of an eigenvalue is a **frequency** and its measure unit is:

$$Hz = s^{-1} = \frac{1}{s}$$

For this reason, eigenvalues are called **natural frequencies** or **modes** of the system

Notes

Dominant Eigenvalue and Transient Duration

Let's consider a asymptotically stable system with eigenvalue:

$$\lambda_1, \lambda_2, ..., \sigma_1 \pm i\omega_1, \sigma_2 \pm i\omega_2, ...$$

The real parts contribute to the response as an exponential term:

$$e^{\lambda_i t} \qquad e^{\sigma_i t}$$

The inverse-absolute of a real part of an eigenvalue is called **time constant**

$$T_i = \frac{1}{|\lambda_i|} \qquad T_i = \frac{1}{|\sigma_i|}$$

Given that λ_* is the dominant eigenvalue (real part), the duration of the transient (set-up time) is approximately:

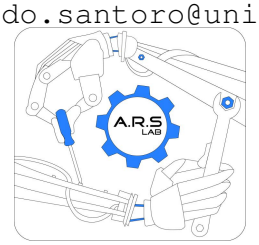
$$T_A \simeq \frac{3}{|\lambda_*|} \qquad T_A \simeq \frac{4}{|\lambda_*|}$$

Notes

A Recap on Dynamic Systems

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Advanced Robotics and Autonomous Systems

Notes